

CASCADE-CLASSICAL CORRESPONDENCE III: REFINEMENT SPACES AND SYMMETRY ACTIONS

ABSTRACT. This paper builds the finite-level analytic infrastructure intended for downstream cascade-correspondence papers. From the algebraic substrate of paper P2, we construct: iterated refinement data $\mathcal{R}_n^\bullet = (V_n^\bullet, E_n^\bullet, F_n^\bullet, \partial_n^\bullet)$ for $\bullet \in \{H_4, D_4\}$, including explicit 2-face data needed to place centroid vertices, with literal vertex inclusion $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$; finite-level ℓ^2 vertex spaces X_n^0 and edge ℓ^2 spaces X_n^1 under counting measure; downward bonding maps $p_{n+1,n}^0$ defined as restriction to $V(G_n^\bullet)$ and $p_{n+1,n}^1$ defined as fibrewise averaging over the canonical edge-parent map; the projective inverse-limit Hilbert space (in the vertex sector) $X_{\text{inv}}^{0,\bullet} \cong \ell^2(V_\infty^\bullet; F^0)$; the cylindrical inductive-limit space $X_{\text{cyl}}^{0,\bullet} = c_{00}(V_\infty^\bullet; F^0)$ embedded densely in $X_{\text{inv}}^{0,\bullet}$; finite-level Coxeter intertwining identities for the bonding maps; the mixed-form vertex-sector σ intertwining identity for the bonding maps; and a restricted Coxeter- σ commutation identity (full $W(H_4)$ - σ commutation on the V_{H_4} sector is not delivered by the source theorem and is not claimed here, since $W(H_4)$ acts via $\mathbb{Q}(\varphi)$ -linear matrices outside the hypothesis of [1, Theorem `thm:pisigma-functorial`]; see Theorem 8.5(4) and Remark 8.6 for the precise scope).

This is the third of thirteen planned cascade infrastructure papers. The construction is in the correct projective direction: bonding maps $p_{n+1,n}$ go *downward* from level $n+1$ to level n , implemented as restriction (vertex sector) or as canonical parent-fibre averaging (edge sector). Their adjoints $j_{n,n+1} := p_{n+1,n}^*$ are (rescaled) embeddings going upward. In the vertex sector the adjoint is a genuine section ($p^0 j^0 = \text{id}$), the cylindrical direct limit is well-defined, and the inverse limit is its Hilbert completion. In the edge sector the adjoint is only a *half-section* ($p^1 j^1 = \frac{1}{2} \text{id}$), a clean cylindrical direct-limit / Hilbert-completion package is not established here, and edge-sector statements are kept at the finite level with bounded bonding maps only (Remarks 6.3 and 7.5). *No continuum-limit, Gromov-Hausdorff, spectral-convergence, or manifold-identification claim is made here.*

The main theorems proved in this paper: (1) Proposition 2.8, the refinement rule with $V_n \subset V_{n+1}$ inclusion is well-defined and Coxeter-equivariant; (2) Proposition 4.3, each finite-level X_n^0, X_n^1 is a finite-dimensional Hilbert space; (3) Theorem 5.3, downward bonding maps are contractions under counting measure; (4) Proposition 5.5, the cylindrical embeddings $j_{n,n+1}$ are the adjoints of the bonding maps; (5) Theorem 6.2, the vertex algebraic direct limit $X_{\text{cyl}}^{0,\bullet}$ is well-defined as $c_{00}(V_\infty^\bullet; F^0)$; (6) Theorem 7.3, the vertex inverse limit $X_{\text{inv}}^{0,\bullet}$ under restriction is canonically isometric to $\ell^2(V_\infty^\bullet; F^0)$ and is a separable Hilbert space; (7) Proposition 7.4, $X_{\text{cyl}}^{0,\bullet}$ embeds densely in $X_{\text{inv}}^{0,\bullet}$; (8) Theorem 8.5, the finite-level vertex Coxeter intertwining $g \circ p_{n+1,n}^0 = p_{n+1,n}^0 \circ g$ and $g \circ j_{n,n+1}^0 = j_{n,n+1}^0 \circ g$ (which descend to $X_{\text{cyl}}^{0,\bullet}$ and $X_{\text{inv}}^{0,\bullet}$), the finite-level edge Coxeter intertwining for p^1 and j^1 , and the mixed-form vertex-sector σ intertwining for p^0 and j^0 on $X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ (using σ from P1).

These are the intended load-bearing inputs for the closure-dynamics paper P4, the metric projection paper P5, and the gauge/arithmetical/computation branch papers P9-P13, as well as for the correspondence Foundations paper.

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1. INTRODUCTION AND SOURCE AUDIT

1.1. The repair target. An earlier draft of the correspondence Foundations manuscript defined the cascade limit Hilbert space using forward zero-extension maps $X_n \rightarrow X_{n+1}$ and called the resulting object an “inverse limit”. That construction is incorrect on the following count: the system as it stands is inductive (not projective). Under the counting-measure ℓ^2 structure used in this paper, zero-extension is in fact isometric (and is the adjoint $j_{n,n+1}^0$ of the restriction-bonding map; see Proposition 5.5 and Lemma 6.5), but the natural inverse limit of the X_n system is built from the projective-direction (downward) bonding maps, not from zero-extension.

The correct construction is built on *downward* bonding maps $p_{n+1,n}: X_{n+1} \rightarrow X_n$. The cleanest mathematical choice for these bonding maps, used here, is:

- Vertex sector: $p_{n+1,n}^0$ is restriction of a function on $V(G_{n+1}^\bullet)$ to its sub-domain $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$.
- Edge sector: $p_{n+1,n}^1$ is fibrewise averaging along the canonical edge-parent map $E^{\text{or}}(G_{n+1}^\bullet) \rightarrow E^{\text{or}}(G_n^\bullet) \cup \{\perp\}$, defined on subdividing edges (which are sent to the level- n oriented edge they are a piece of) and assigning $\perp$ to all other classes (so the averaging is supported on subdividing-edge fibres only).

Both maps are genuine contractions under *counting measure* on finite vertex/edge sets, and their adjoints are the natural embeddings in the upward direction. The projective system in the correct direction is then $X_{\text{inv}} := \{(f_n) : p_{n+1,n}f_{n+1} = f_n\}$, equipped with the limit inner product, and we show this is canonically isometric to $\ell^2(V_\infty^\bullet)$ for the countable union $V_\infty^\bullet := \bigcup_n V(G_n^\bullet)$.

1.2. What this paper proves. The substrate constructed here consists of:

- (1) Iterated refinement graphs $G_n^{H_4}$ and $G_n^{D_4}$ with $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ and a canonical edge-parent map (§2).

- (2) Counting measures on vertices and oriented edges (§3).
- (3) Finite-level vertex and edge ℓ^2 spaces X_n^0, X_n^1 , each a finite-dimensional Hilbert space (§4).
- (4) Downward bonding maps $p_{n+1,n}^0$ (restriction) and $p_{n+1,n}^1$ (parent-fibre averaging), both contractions (§5).
- (5) Cylindrical embeddings $j_{n,n+1}^0$ (zero-extension) and $j_{n,n+1}^1$ (rescaled replication) as adjoints of the bonding maps (§5).
- (6) Algebraic direct-limit space (vertex sector) $X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0)$ (§6).
- (7) Projective inverse-limit Hilbert space (vertex sector) $X_{\text{inv}}^{0,\bullet} \cong \ell^2(V_\infty^\bullet; F^0)$ (§7).
- (8) Finite-level Coxeter intertwining (vertex and edge, for p and j) and mixed-form vertex σ intertwining (for p^0 and j^0) (§8).

1.3. What this paper does not prove. P3 is deliberately narrow. We record the exclusions explicitly:

- (1) No continuum-limit, Gromov–Hausdorff, or spectral-convergence statement.
- (2) No closure-functional, gradient-flow, energy, or monotonicity theorem.
- (3) No identification of X_{inv} with any classical continuum function space such as $L^2(S^3)$.
- (4) No metric, hydrodynamic, gauge, spectral, zeta, Hecke, or CTM/TM correspondence theorem.
- (5) No vertex-wise quotient map $V(G_n^{H_4}) \rightarrow V(G_n^{D_4})$.
- (6) No physical or phenomenological interpretation.

1.4. Source audit. Theorem inputs to this paper are:

Internal cascade infrastructure.: Paper P1 (cascade–classical correspondence I, σ -fixed rationality [1]) for the coefficientwise σ -action; paper P2 (cascade algebraic substrate [2]) for the algebraic substrate, including $D_4 \subset E_8$, the icosian model of H_4 , the explicit V_{600} and V_{24} vertex sets, and the $\text{Cl}(1,3)$ and octonionic sector models.

Classical mathematics.: Standard Hilbert space and direct/inverse limit theory [7, 8]; standard Coxeter group theory [9, 10]; finite graph and 1-cochain conventions [11]; elementary ℓ^2 theory.

We do not cite as theorem-grade input any other cascade-internal note. The bibliography below distinguishes load-bearing upstream sources (P1, P2) used as theorem-grade input from downstream cascade preprints ([4, 5, 6]) that consume the present paper’s results, and classical references used as background or auxiliary input.

1.5. Organization. Section 2 sets up the refinement graphs and the canonical edge-parent map. Section 3 fixes the counting measures. Section 4 defines the finite-level vertex and edge ℓ^2 spaces. Section 5 constructs the downward bonding maps and their adjoints, the cylindrical embeddings. Section 6 treats the cylindrical algebraic direct-limit space. Section 7 constructs the projective limit Hilbert space and proves it is $\ell^2(V_\infty)$. Section 8 establishes the intertwining identities for refinement, Coxeter, and σ . Section 9 states the citation contract for downstream papers. Appendix A is a notation concordance.

2. REFINEMENT GRAPHS AND THE EDGE-PARENT MAP

2.1. The base graphs.

Definition 2.1 (Base graphs). Let $V_{600} \subset \mathbb{H} \cong \mathbb{R}^4$ be the standard 120-vertex set of the regular 600-cell, in the coordinate realization of [2, Definition `def:V600`]. Let $V_{24} \subset V_{600}$ be the 24-vertex subset (axis points and half-type points) congruent to the rescaled D_4 root polytope by [2, Definition `def:V24` and Proposition `prop:24-in-600`].

The *base H_4 graph* $G_0^{H_4} = (V_0^{H_4}, E_0^{H_4})$ has vertex set $V_0^{H_4} := V_{600}$ and unoriented edges joining pairs of 600-cell vertices at the standard 600-cell edge distance [10, §22.7].

The *base D_4 graph* $G_0^{D_4} = (V_0^{D_4}, E_0^{D_4})$ has vertex set $V_0^{D_4} := V_{24}$ and unoriented edges joining pairs of 24-cell vertices at the standard 24-cell edge distance [10, §22.2].

We write $\bullet$ for either decoration and $G_n^\bullet$ for the level- n graph below. The Coxeter group $W^\bullet$ is $W(H_4)$ when $\bullet = H_4$ and $W(D_4)$ when $\bullet = D_4$.

2.2. Refinement with vertex inclusion. A graph alone does not determine the Schläfli refinement; the construction needs explicit 2-face data, since centroids are indexed by 2-faces. We carry that data through levels as a *refinement datum*.

Definition 2.2 (Refinement datum). A *refinement datum at level n* is a tuple $\mathcal{R}_n^\bullet = (V_n^\bullet, E_n^\bullet, F_n^\bullet, \partial_n^\bullet)$ where $V_n^\bullet$ and $E_n^\bullet$ are the vertex and unoriented edge sets of $G_n^\bullet$; $F_n^\bullet$ is a finite set of 2-faces (each face is a finite cyclic sequence of edges $(e_1, \dots, e_k) \in E_n^\bullet$ with $k \geq 3$ closing up into a polygon, with the standard cyclic-equivalence convention); and $\partial_n^\bullet: F_n^\bullet \rightarrow \bigsqcup_k (E_n^\bullet)^k$ is the boundary map sending each face to its edge sequence. We require only that every face boundary sits in $E_n^\bullet$ (i.e. $\partial_n^\bullet(f) \in (E_n^\bullet)^*$ for every $f \in F_n^\bullet$); we do *not* require that every edge belong to some face boundary. This is what we mean by “graph-plus-faces datum” as opposed to a 2-dimensional cell complex, and is the convention used at every refined level by Definition 2.3 below. The base data $\mathcal{R}_0^\bullet$ is induced by Definition 2.1 from the 1-skeleton and 2-faces of the underlying 4-polytope (V_{600} for $\bullet = H_4$, V_{24} for $\bullet = D_4$). $W^\bullet$ acts on each component of $\mathcal{R}_n^\bullet$ compatibly with the boundary map.

Definition 2.3 (Schläfli-style refinement step with vertex inclusion). Given a refinement datum $\mathcal{R}_n^\bullet = (V_n^\bullet, E_n^\bullet, F_n^\bullet, \partial_n^\bullet)$, the *Schläfli-style refinement step* produces a new datum $\mathcal{R}_{n+1}^\bullet$ in the sense of Definition 2.2 with the following explicit components. Throughout, $E_{n+1}^\bullet$ is a *labelled-edge multiset*: each edge generated below is tagged by its class $\in \{(a), (b), (c)\}$ and its parent face f (when applicable), so two edges with the same unordered endpoint pair but different tags are distinct elements of $E_{n+1}^\bullet$. (This convention is included so that the construction is well-defined regardless of whether distinct parent faces happen to generate the same unordered endpoint pair; under it, no claim or computation below depends on whether that coincidence occurs.)

- **Vertices.** $V_{n+1}^\bullet := V_n^\bullet \sqcup M_n^\bullet \sqcup C_n^\bullet$, where $M_n^\bullet$ contains exactly one edge-midpoint m_e per $e \in E_n^\bullet$, and $C_n^\bullet$ contains exactly one face-centroid c_f per $f \in F_n^\bullet$. The first summand is the literal inclusion $V_n^\bullet \subset V_{n+1}^\bullet$.
- **Edges.** $E_{n+1}^\bullet$ is the disjoint union of three explicit edge classes:
 - (a) *Subdividing edges*: for each $e = \{u, v\} \in E_n^\bullet$, the two edges $\{u, m_e\}$ and $\{m_e, v\}$.
 - (b) *Midpoint-centroid edges*: for each face $f \in F_n^\bullet$ with edge sequence $(e_1, \dots, e_k) = \partial_n^\bullet(f)$, the k edges $\{m_{e_i}, c_f\}$ for $i = 1, \dots, k$.
 - (c) *Midpoint-midpoint diagonal edges*: for each face $f \in F_n^\bullet$ with edge sequence $(e_1, \dots, e_k)$, the k edges $\{m_{e_i}, m_{e_{i+1}}\}$ (cyclic indices), closing each level- $(n+1)$ triangle face below.
- **Faces.** $F_{n+1}^\bullet$ is the disjoint union of level- $(n+1)$ triangles: each level- n face f with edge sequence $(e_1, \dots, e_k)$ produces k triangles $(m_{e_i}, c_f, m_{e_{i+1}})$ (cyclic indices), each with edge sequence $(\{m_{e_i}, c_f\}, \{c_f, m_{e_{i+1}}\}, \{m_{e_{i+1}}, m_{e_i}\})$ drawn from edge classes (b), (b), (c) respectively. All three boundary edges lie in $E_{n+1}^\bullet$ by construction, so $\partial_{n+1}^\bullet$ is well-defined and the iterated refinement datum is well-formed in the sense of Definition 2.2 (a graph-plus-faces datum in which every face boundary lies in the edge set, but no converse incidence is required: an edge need not belong to any face boundary). In particular, the class-(a) subdividing edges $\{u, m_e\}, \{m_e, v\}$ are present in $E_{n+1}^\bullet$ but do not appear in any $\partial_{n+1}^\bullet(f)$; this is intentional and is the only place the datum departs from a classical 2-dimensional cell complex (in which one would retain the 2-cells of f as 2-cells of the next level rather than subdividing them into triangles).
- **$W^\bullet$ -action.** $W^\bullet$ acts on $V_{n+1}^\bullet$ by $g \cdot v = gv$ for $v \in V_n^\bullet$, $g \cdot m_e = m_{ge}$ for $m_e \in M_n^\bullet$ (well-defined since $g\{u, v\} = \{gu, gv\}$), and $g \cdot c_f = c_{gf}$ for $c_f \in C_n^\bullet$ (well-defined since $W^\bullet$ acts on $F_n^\bullet$). The induced action on $E_{n+1}^\bullet$ permutes each of the three edge classes (a), (b), (c) among themselves and on $F_{n+1}^\bullet$ permutes triangles among themselves.

Remark 2.4 (Scope of the construction). The chain-space construction below uses the full edge set $E_{n+1}^\bullet$ (all three classes (a), (b), (c)) as the domain of $X_{n+1}^{1, \bullet}$. The edge bonding map

$p_{n+1,n}^1$ of Definition 5.2 (defined via the edge-parent map of Definition 2.6) averages only over class-(a) preimages (the subdividing edges); classes (b) and (c) and the centroid vertices in $V_{n+1}^\bullet$ are present in the level- $(n+1)$ data but are ignored by the parent-fibre averaging at this step. The level- $(n+1)$ face structure $F_{n+1}^\bullet$ is carried only to keep Definition 2.2 valid for iteration; no construction below depends on $F_{n+1}^\bullet$. Downstream papers [4, 5] either inherit the full datum or use a deliberately reduced “pure-midpoint” abstract model that omits the face structure; [5, §sec:scope-and-gap] names the gap explicitly.

Remark 2.5 (Vertex inclusion is the key). The literal subset $V_n^\bullet \subset V_{n+1}^\bullet$ from Definition 2.3 is the structural feature that makes the entire bonding-map construction below work cleanly: restriction of a function on $V_{n+1}^\bullet$ to $V_n^\bullet$ is a well-defined function on $V_n^\bullet$, and (under counting measure) is automatically a contraction in the ℓ^2 -norm.

2.3. The canonical edge-parent map. Unlike vertices, edges are not preserved under refinement: each level- n edge $\{u, v\} \in E_n^\bullet$ is replaced by two level- $(n+1)$ edges $\{u, m\}$ and $\{m, v\}$. We record the inverse direction as a function:

Definition 2.6 (Edge-parent map). For oriented edges, define

$$\pi_{n+1,n}^E: E^{\text{or}}(G_{n+1}^\bullet) \rightarrow E^{\text{or}}(G_n^\bullet) \cup \{\perp\}$$

by:

- If $e' \in E^{\text{or}}(G_{n+1}^\bullet)$ is one of the two pieces of a subdivided level- n oriented edge $e = (u, v) \in E^{\text{or}}(G_n^\bullet)$ — i.e., $e' = (u, m)$ or $e' = (m, v)$ where m is the midpoint of $\{u, v\}$ — then $\pi_{n+1,n}^E(e') := e$ (with the orientation inherited).
- If e' is one of the new level- $(n+1)$ edges connecting a midpoint to a centroid or two midpoints (i.e., a “boundary” edge introduced by Schläfli refinement that does not subdivide a level- n edge), then $\pi_{n+1,n}^E(e') := \perp$.

The edge-parent map is well-defined as a function and is $W^\bullet$ -equivariant: $g \cdot \pi_{n+1,n}^E(e') = \pi_{n+1,n}^E(g \cdot e')$ for every $g \in W^\bullet$ and every $e' \in E^{\text{or}}(G_{n+1}^\bullet)$. This holds because Schläfli refinement is itself $W^\bullet$ -equivariant, so $W^\bullet$ permutes the descendants of an edge among themselves.

Notation 2.7 (Subdividing edges). We write $E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet) := \{e' : \pi_{n+1,n}^E(e') \neq \perp\}$ for the set of level- $(n+1)$ oriented edges that subdivide a level- n edge. By construction, $|E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet)| = 2 \cdot |E^{\text{or}}(G_n^\bullet)|$ (each level- n edge contributes exactly two level- $(n+1)$ subdividing edges).

2.4. Properties of the refinement.

Proposition 2.8 (Refinement properties). *For every $n \geq 0$ and $\bullet \in \{H_4, D_4\}$:*

- (1) $V_n^\bullet \subset V_{n+1}^\bullet$ is a literal subset inclusion of finite sets.
- (2) The action of $W^\bullet$ on $V_n^\bullet$ extends to an action on $V_{n+1}^\bullet$ that commutes with the inclusion.
- (3) The edge-parent map $\pi_{n+1,n}^E$ is well-defined and $W^\bullet$ -equivariant.
- (4) Each level- n oriented edge has exactly two level- $(n+1)$ preimages under $\pi_{n+1,n}^E$ (its two subdividing pieces in the inherited orientation).

Proof. (1) and (2) follow directly from Definition 2.3: the vertex inclusion is by construction, and the $W^\bullet$ -action extends because midpoints and centroids are $W^\bullet$ -equivariantly defined from the level- n data.

(3) Edge-parent equivariance: by Definition 2.3 and the equivariant action on labelled descendants (the $W^\bullet$ -action permutes labelled level- $(n+1)$ edges among themselves, preserving the class-label and the parent face), so the edge-parent map respects this permutation.

(4) Each level- n unoriented edge $\{u, v\}$ contributes oriented edges (u, v) and (v, u) at level n , each of which has two subdividing pieces at level $n+1$ (namely $(u, m), (m, v)$ for (u, v) , and $(v, m), (m, u)$ for (v, u)). So there are two preimages per oriented edge, giving the $|E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet)| = 2|E^{\text{or}}(G_n^\bullet)|$ count of Notation 2.7. $\square$

Remark 2.9 (No non-canonical choices). Definition 2.6 introduces no non-canonical choice: each level- $(n+1)$ oriented edge is either intrinsically a subdividing piece of a unique level- n

oriented edge (in which case it has a unique parent), or it is intrinsically a boundary edge of the Schläfli refinement (in which case it has parent $\perp$). No lex-order, evaluation, or vertex-choice rule is needed.

3. COUNTING MEASURES

Definition 3.1 (Counting measures). For each $n \geq 0$ and $\bullet \in \{H_4, D_4\}$:

$$\begin{aligned}\mu_n^\bullet(S) &:= |S|, & S \subseteq V(G_n^\bullet); \\ \nu_n^\bullet(T) &:= |T|, & T \subseteq E^{\text{or}}(G_n^\bullet).\end{aligned}$$

These are unweighted counting measures; in particular, every vertex and oriented edge has measure 1.

Remark 3.2 (Why counting measure). We use unscaled counting measure rather than probability normalization. The reason is that, with $V_n^\bullet \subset V_{n+1}^\bullet$, restriction to $V_n^\bullet$ is a contraction in $\ell^2(V_{n+1}^\bullet, \mu_{n+1}^\bullet) \rightarrow \ell^2(V_n^\bullet, \mu_n^\bullet)$ *automatically*: the norm of the restriction is $\sum_{v \in V_n^\bullet} |f(v)|^2$, which is at most $\sum_{w \in V_{n+1}^\bullet} |f(w)|^2 = \|f\|_{X_{n+1}^0}^2$ since the smaller sum has fewer terms (all nonneg). Probability normalization would introduce a level-dependent rescaling factor that breaks this property.

Notation 3.3 (Sector fibres). We import the following sector fibres and inner products from [2]:

- $V_{H_4} \cong \mathbb{H} \cong \mathbb{R}^4$ with the standard Euclidean inner product (square-norm of unit icosian = 1), via [2, Definitions `def:icosian-ring`, `def:V600`];
- $V_{D_4} \cong \mathbb{R}^4$ with the standard Euclidean inner product, via [2, Definition `def:V24` and Proposition `prop:24-in-600`];
- $V_{16} = \text{Cl}(1, 3)$ as a real 16-dimensional vector space with the positive-definite Euclidean inner product $\langle \cdot, \cdot \rangle_{\text{Cl}}$ defined by declaring the monomial basis $\{\gamma_A : A \subseteq \{0, 1, 2, 3\}\}$ from [2, Definition `def:Cl-basis`] orthonormal: for $a = \sum_A a_A \gamma_A$, $b = \sum_A b_A \gamma_A$ with $a_A, b_A \in \mathbb{R}$,

$$\langle a, b \rangle_{\text{Cl}} := \sum_{A \subseteq \{0, 1, 2, 3\}} a_A b_A.$$

Equivalently this is the Hilbert–Schmidt-style trace form $\langle \tilde{a}^* b \rangle_0$ of [12, Ch. 9], with $\tilde{a}$ the reversion and a^* the Clifford conjugation defined by $\gamma_0^* = \gamma_0$, $\gamma_i^* = -\gamma_i$ for $i \in \{1, 2, 3\}$ (so that $\gamma_A^* \gamma_A = 1$ for every A), and $\langle \cdot \rangle_0$ the scalar-grade projection; positivity on the monomial basis is the orthonormality just stated, hence the form is positive-definite on V_{16} ;

- $V_8 = \mathbb{O}$ with the standard octonion norm form $\langle a, b \rangle_{\mathbb{O}} := \Re(a\bar{b})$, via [2, Definition `def:octonions`].

We fix the combined vertex fibre

$$F^0 := V_{H_4} \oplus V_{D_4} \oplus V_{16} \oplus V_8,$$

$\mathbb{R}$ -dimension $4 + 4 + 16 + 8 = 32$, with the orthogonal-direct-sum inner product, and the edge fibre

$$F^1 := \text{Im}(\mathbb{O}),$$

$\mathbb{R}$ -dimension 7, with the Euclidean inner product induced from $\mathbb{O}$. These choices are fixed throughout this paper. (Downstream papers needing a different edge fibre re-construct the relevant spaces from scratch; we make no template promise to accommodate arbitrary F^1 .)

4. SECTOR FIBRES AND FINITE-LEVEL ℓ^2 SPACES

Definition 4.1 (Vertex-sector ℓ^2 space). For $n \geq 0$ and $\bullet \in \{H_4, D_4\}$, define

$$X_n^{0, \bullet} := \ell^2(V(G_n^\bullet), \mu_n^\bullet; F^0),$$

the F^0 -valued ℓ^2 -functions on $V(G_n^\bullet)$ under counting measure, with inner product

$$\langle f, g \rangle_{X_n^0} := \sum_{v \in V(G_n^\bullet)} \langle f(v), g(v) \rangle_{F^0}.$$

Definition 4.2 (Edge-sector ℓ^2 space). For $n \geq 0$ and $\bullet \in \{H_4, D_4\}$, define

$$X_n^{1,\bullet} := \ell_{\text{anti}}^2(E^{\text{or}}(G_n^\bullet), \nu_n^\bullet; F^1),$$

the F^1 -valued ℓ^2 -functions A on oriented edges satisfying the antisymmetry convention $A(-e) = -A(e)$ for every oriented edge e (where $-e$ denotes reversed orientation), with inner product

$$\langle A, B \rangle_{X_n^1} := \frac{1}{2} \sum_{e \in E^{\text{or}}(G_n^\bullet)} \langle A(e), B(e) \rangle_{F^1}.$$

The factor of $\frac{1}{2}$ accounts for the double-counting from including both orientations of each unoriented edge.

Proposition 4.3 (Finite-level Hilbert structure). *For every $n \geq 0$ and $\bullet \in \{H_4, D_4\}$:*

- (1) $X_n^{0,\bullet}$ and $X_n^{1,\bullet}$ are finite-dimensional real Hilbert spaces.
- (2) $\dim_{\mathbb{R}} X_n^{0,\bullet} = |V(G_n^\bullet)| \cdot 32$ and $\dim_{\mathbb{R}} X_n^{1,\bullet} = |E(G_n^\bullet)| \cdot 7$, where $|V(G_n^\bullet)|$ counts elements of the vertex set $V_n^\bullet$ and $|E(G_n^\bullet)|$ counts labelled unoriented edge elements of $E_n^\bullet$ (Definition 2.3; in particular tagged duplicates are counted separately), with $|E(G_n^\bullet)| = |E^{\text{or}}(G_n^\bullet)|/2$.
- (3) In particular, both spaces are separable Hilbert spaces.

Proof. Each fibre is a finite-dimensional Euclidean space (32 for F^0 , 7 for F^1). The vertex set $V_n^\bullet$, the unoriented edge set $E_n^\bullet$ (as a labelled multiset, per Definition 2.3), and the face set $F_n^\bullet$ are all finite by induction from the finite base data $\mathcal{R}_0^\bullet$ (Definitions 2.1, 2.2): each refinement step produces $|M_n^\bullet| = |E_n^\bullet|$ midpoints, $|C_n^\bullet| = |F_n^\bullet|$ centroids, a number of new edges bounded by $2|E_n^\bullet| + \sum_{f \in F_n^\bullet} 2 \cdot \text{len}(\partial_n^\bullet f)$ (classes (a)+(b)+(c)), and a number of new faces $\sum_{f \in F_n^\bullet} \text{len}(\partial_n^\bullet f)$, all finite. Hence $E^{\text{or}}(G_n^\bullet)$ is finite. A finite direct sum of finite-dimensional Euclidean spaces under a positive-definite bilinear form is a finite-dimensional Hilbert space [7, Ch. II]. The dimension formulas follow by counting; the antisymmetry convention identifies each pair $(e, -e)$ with a single element of dimension $\dim_{\mathbb{R}} F^1 = 7$, giving the unoriented-edge count. Separability is automatic. $\square$

5. DOWNWARD BONDING MAPS AND THEIR ADJOINTS

5.1. Vertex-sector bonding by restriction.

Definition 5.1 (Vertex bonding map). For $n \geq 0$, define the *vertex bonding map*

$$p_{n+1,n}^0: X_{n+1}^0 \rightarrow X_n^0, \quad (p_{n+1,n}^0 f)(v) := f(v) \quad \text{for } v \in V(G_n^\bullet).$$

Since $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ (Definition 2.3), the right-hand side is well-defined: f is already a function on $V(G_{n+1}^\bullet)$ and we simply restrict its domain to the subset $V(G_n^\bullet)$.

5.2. Edge-sector bonding by parent-fibre averaging.

Definition 5.2 (Edge bonding map). For $n \geq 0$, define the *edge bonding map*

$$p_{n+1,n}^1: X_{n+1}^1 \rightarrow X_n^1, \quad (p_{n+1,n}^1 A)(e) := \frac{1}{2} \sum_{e' \in (\pi_{n+1,n}^E)^{-1}(e)} A(e')$$

for $e \in E^{\text{or}}(G_n^\bullet)$. By Proposition 2.8(4), the parent fibre $(\pi_{n+1,n}^E)^{-1}(e)$ has exactly two oriented edges (the two subdividing pieces of e), so the sum has two terms and the average factor $\frac{1}{2}$ gives the standard arithmetic mean.

We verify that $p_{n+1,n}^1 A$ satisfies the antisymmetry convention: $(p_{n+1,n}^1 A)(-e) = \frac{1}{2} \sum A(-e') = -\frac{1}{2} \sum A(e') = -(p_{n+1,n}^1 A)(e)$, since each subdividing piece of $-e$ is the reverse of a subdividing piece of e .

5.3. Bonding maps are contractions.

Theorem 5.3 (Bonding contractions). *For every $n \geq 0$:*

- (1) $p_{n+1,n}^0: X_{n+1}^0 \rightarrow X_n^0$ is a contraction: $\|p_{n+1,n}^0 f\|_{X_n^0} \leq \|f\|_{X_{n+1}^0}$ for all $f \in X_{n+1}^0$.
- (2) $p_{n+1,n}^1: X_{n+1}^1 \rightarrow X_n^1$ is a contraction: $\|p_{n+1,n}^1 A\|_{X_n^1} \leq \|A\|_{X_{n+1}^1}$ for all $A \in X_{n+1}^1$.

Proof. (1) By Definition 5.1, $\|p_{n+1,n}^0 f\|_{X_n^0}^2 = \sum_{v \in V(G_n^\bullet)} \|f(v)\|_{F^0}^2$. Since $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ and $\|f(v)\|_{F^0}^2 \geq 0$ for every v ,

$$\|p_{n+1,n}^0 f\|_{X_n^0}^2 = \sum_{v \in V(G_n^\bullet)} \|f(v)\|_{F^0}^2 \leq \sum_{w \in V(G_{n+1}^\bullet)} \|f(w)\|_{F^0}^2 = \|f\|_{X_{n+1}^0}^2.$$

(2) By Definition 5.2 and the Cauchy–Schwarz inequality applied to the two-term average $\frac{1}{2}(A(e_1) + A(e_2))$:

$$\frac{1}{2}\|(p_{n+1,n}^1 A)(e)\|_{F^1}^2 = \frac{1}{2} \cdot \frac{1}{4} \|A(e_1) + A(e_2)\|_{F^1}^2 \leq \frac{1}{2} \cdot \frac{1}{2} (\|A(e_1)\|^2 + \|A(e_2)\|^2),$$

where the first equality is the antisymmetry convention's $\frac{1}{2}$ factor and the inequality is Cauchy–Schwarz on the two-term average. Summing over $e \in E^{\text{or}}(G_n^\bullet)$ and using that each subdividing piece e' of level $n+1$ contributes to exactly one parent (by Proposition 2.8(4)):

$$\|p_{n+1,n}^1 A\|_{X_n^1}^2 \leq \frac{1}{2} \sum_{e' \in E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet)} \|A(e')\|_{F^1}^2 \leq \frac{1}{2} \sum_{e' \in E^{\text{or}}(G_{n+1}^\bullet)} \|A(e')\|_{F^1}^2 = \|A\|_{X_{n+1}^1}^2,$$

where the final inequality is because $E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet) \subseteq E^{\text{or}}(G_{n+1}^\bullet)$. (The Cauchy–Schwarz step gives the sharper coefficient $\frac{1}{4}$ on the subdividing-edge sum, but the weaker coefficient $\frac{1}{2}$ used here suffices for the contraction bound and is what we cite below.) $\square$

5.4. Adjoint cylindrical embeddings.

Definition 5.4 (Cylindrical embeddings). For $n \geq 0$, define:

- *Vertex embedding* $j_{n,n+1}^0: X_n^0 \rightarrow X_{n+1}^0$ by zero-extension:

$$(j_{n,n+1}^0 g)(w) := \begin{cases} g(w) & \text{if } w \in V(G_n^\bullet), \\ 0 & \text{if } w \in V(G_{n+1}^\bullet) \setminus V(G_n^\bullet). \end{cases}$$

- *Edge embedding (adjoint of p^1)* $j_{n,n+1}^1: X_n^1 \rightarrow X_{n+1}^1$ by rescaled parent-replication:

$$(j_{n,n+1}^1 B)(e') := \begin{cases} \frac{1}{2} B(\pi_{n+1,n}^E(e')) & \text{if } \pi_{n+1,n}^E(e') \neq \perp, \\ 0 & \text{otherwise.} \end{cases}$$

The rescaling factor $\frac{1}{2}$ is required for $j_{n,n+1}^1$ to be the genuine ℓ^2 -adjoint of $p_{n+1,n}^1$ (Proposition 5.5(3) below). Without this factor, the unscaled replication $s_{n,n+1}^1 B(e') := B(\pi^E(e'))$ is a *section* of $p_{n+1,n}^1$ (i.e. $p_{n+1,n}^1 \circ s_{n,n+1}^1 = \text{id}$, Remark 5.6 below) but not its adjoint.

Proposition 5.5 (Adjoint relation). *For every $n \geq 0$:*

- (1) $j_{n,n+1}^0 = (p_{n+1,n}^0)^*$ as bounded linear maps, i.e. $\langle p_{n+1,n}^0 f, g \rangle_{X_n^0} = \langle f, j_{n,n+1}^0 g \rangle_{X_{n+1}^0}$ for all $f \in X_{n+1}^0, g \in X_n^0$.
- (2) $p_{n+1,n}^0 \circ j_{n,n+1}^0 = \text{id}_{X_n^0}$ (vertex embedding is a section).
- (3) $j_{n,n+1}^1 = (p_{n+1,n}^1)^*$ as bounded linear maps.
- (4) $p_{n+1,n}^1 \circ j_{n,n+1}^1 = \frac{1}{2} \text{id}_{X_n^1}$ (the genuine adjoint j^1 is a half-section, not an exact section, of p^1 ; see also Remark 5.6).

Proof. (1)

$$\begin{aligned}
 \langle p_{n+1,n}^0 f, g \rangle_{X_n^0} &= \sum_{v \in V(G_n^\bullet)} \langle f(v), g(v) \rangle_{F^0} \\
 &= \sum_{w \in V(G_{n+1}^\bullet)} \langle f(w), (j_{n,n+1}^0 g)(w) \rangle_{F^0} \quad (\text{extending by zero}) \\
 &= \langle f, j_{n,n+1}^0 g \rangle_{X_{n+1}^0}.
 \end{aligned}$$

(2) For $g \in X_n^0$ and $v \in V(G_n^\bullet)$, $(p_{n+1,n}^0 j_{n,n+1}^0 g)(v) = (j_{n,n+1}^0 g)(v) = g(v)$.

(3) Compute both sides explicitly. With the edge-sector inner product factor of $\frac{1}{2}$ from Definition 4.2:

$$\begin{aligned}
 \langle p_{n+1,n}^1 A, B \rangle_{X_n^1} &= \frac{1}{2} \sum_{e \in E^{\text{or}}(G_n^\bullet)} \langle (p_{n+1,n}^1 A)(e), B(e) \rangle_{F^1} \\
 &= \frac{1}{2} \sum_e \frac{1}{2} \sum_{e' \in (\pi^E)^{-1}(e)} \langle A(e'), B(e) \rangle_{F^1} \\
 &= \frac{1}{4} \sum_{e' \in E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet)} \langle A(e'), B(\pi^E(e')) \rangle_{F^1}.
 \end{aligned}$$

With the rescaled $j_{n,n+1}^1$ formula $(j_{n,n+1}^1 B)(e') = \frac{1}{2} B(\pi^E(e'))$ on subdividing edges and 0 elsewhere:

$$\begin{aligned}
 \langle A, j_{n,n+1}^1 B \rangle_{X_{n+1}^1} &= \frac{1}{2} \sum_{e' \in E^{\text{or}}(G_{n+1}^\bullet)} \langle A(e'), (j_{n,n+1}^1 B)(e') \rangle_{F^1} \\
 &= \frac{1}{2} \sum_{e' \in E_{\text{sub}}^{\text{or}}} \langle A(e'), \frac{1}{2} B(\pi^E(e')) \rangle_{F^1} \\
 &= \frac{1}{4} \sum_{e' \in E_{\text{sub}}^{\text{or}}} \langle A(e'), B(\pi^E(e')) \rangle_{F^1}.
 \end{aligned}$$

The two expressions agree, so $j_{n,n+1}^1 = (p_{n+1,n}^1)^*$.

(4) For $B \in X_n^1$ and $e \in E^{\text{or}}(G_n^\bullet)$, the parent fibre of e has two oriented edges e_1, e_2 , both satisfying $\pi^E(e_i) = e$. Thus $(j_{n,n+1}^1 B)(e_i) = \frac{1}{2} B(e)$ for $i = 1, 2$, and $(p_{n+1,n}^1 j_{n,n+1}^1 B)(e) = \frac{1}{2} ((j_{n,n+1}^1 B)(e_1) + (j_{n,n+1}^1 B)(e_2)) = \frac{1}{2} (\frac{1}{2} B(e) + \frac{1}{2} B(e)) = \frac{1}{2} B(e)$. So $p_{n+1,n}^1 \circ j_{n,n+1}^1 = \frac{1}{2} \text{id}_{X_n^1}$. $\square$

Remark 5.6 (Edge section vs adjoint). The edge sector has *two* naturally-related upward maps from X_n^1 to X_{n+1}^1 :

- the *adjoint* $j_{n,n+1}^1 := (p_{n+1,n}^1)^*$, which satisfies $j_{n,n+1}^1 B(e') = \frac{1}{2} B(\pi^E(e'))$ on subdividing edges and $p_{n+1,n}^1 \circ j_{n,n+1}^1 = \frac{1}{2} \text{id}$ (Proposition 5.5(4));
- the *section* $s_{n,n+1}^1 B(e') := B(\pi^E(e'))$ on subdividing edges and 0 elsewhere, which satisfies $p_{n+1,n}^1 \circ s_{n,n+1}^1 = \text{id}$ but is not the ℓ^2 -adjoint of $p_{n+1,n}^1$.

The two differ by the rescaling $j_{n,n+1}^1 = \frac{1}{2} s_{n,n+1}^1$. We use j (the adjoint) below in the finite-level intertwining identities of Theorem 8.5; we use s (the section) only when an exact section is needed downstream. (No edge inverse-limit Hilbert space is constructed in this paper; the inverse-limit construction is purely for the vertex sector.) The vertex sector does not have this distinction because the vertex bonding map is restriction (not averaging), and the corresponding adjoint $j_{n,n+1}^0$ is zero-extension, which is a section ($p^0 \circ j^0 = \text{id}$).

6. THE CYLINDRICAL ALGEBRAIC DIRECT-LIMIT SPACE

6.1. Algebraic direct limit.

Definition 6.1 (Algebraic direct limit on vertices). The *cylindrical algebraic direct-limit space* (vertex sector) is

$$X_{\text{cyl}}^{0,\bullet} := \varinjlim_n (X_n^{0,\bullet}, j_{n,n+1}^0),$$

the direct limit of the system $(X_n^{0,\bullet})_{n \geq 0}$ under the cylindrical embeddings $j_{n,n+1}^0$ of Definition 5.4.

Theorem 6.2 (Identification of $X_{\text{cyl}}^{0,\bullet}$). *The vertex cylindrical direct-limit space identifies canonically with c_{00} -functions on the countable union $V_\infty^\bullet$ of refinement-vertex sets:*

$$X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0),$$

where $V_\infty^\bullet := \bigcup_{n \geq 0} V(G_n^\bullet)$ (a countable set, by Proposition 2.8(1) and the inclusion chain) and $c_{00}(\cdot; F^0)$ denotes F^0 -valued functions of finite support.

Proof. The algebraic direct limit $\varinjlim_n X_n^{0,\bullet}$ is the disjoint union $\bigsqcup_n X_n^{0,\bullet}$ modulo the equivalence relation generated by $f \sim j_{n,n+1}^0 f$ for $f \in X_n^{0,\bullet}$ [7, Ch. II.1.4]. Each equivalence class has a unique representative at the smallest level n for which the class contains an element; that representative is a function on $V(G_n^\bullet)$, which by zero-extension to all higher levels gives a function on $V_\infty^\bullet$ vanishing outside $V(G_n^\bullet)$.

Conversely, every c_{00} -function $\phi: V_\infty^\bullet \rightarrow F^0$ has support contained in $V(G_n^\bullet)$ for some n (since the support is finite and $V_\infty^\bullet$ is the ascending union), so it represents an element of $X_n^{0,\bullet}$.

These maps are inverse to each other and are $\mathbb{R}$ -linear, giving the identification $X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0)$. $\square$

Remark 6.3 (Edge sector: half-section, finite-level only). The edge cylindrical direct-limit construction of the vertex sector does *not* transfer cleanly to edges. The adjoint $j_{n,n+1}^1 = (p_{n+1,n}^1)^*$ satisfies only $p_{n+1,n}^1 \circ j_{n,n+1}^1 = \frac{1}{2} \text{id}_{X_n^1}$ (Proposition 5.5(4)): it is a *half-section*, not a section, of $p_{n+1,n}^1$. Consequently $\varinjlim_n (X_n^{1,\bullet}, j_{n,n+1}^1)$ does not enjoy the properties used in the vertex identification (Theorem 6.2), and we do *not* define an edge cylindrical direct-limit space here. Downstream papers requiring the edge sector should work at the finite level X_n^1 with the bounded bonding maps $p_{n+1,n}^1$ and their adjoints, without relying on a cylindrical-direct-limit formalism for edges.

6.2. The ℓ^2 -norm on X_{cyl} .

Definition 6.4 (Pre-Hilbert structure on X_{cyl}). On $X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0)$, define the inner product

$$\langle \phi, \psi \rangle_{X_{\text{cyl}}^0} := \sum_{v \in V_\infty^\bullet} \langle \phi(v), \psi(v) \rangle_{F^0}.$$

This sum is finite since ϕ, ψ have finite support. The resulting inner product is positive-definite (since $\langle \cdot, \cdot \rangle_{F^0}$ is) and makes X_{cyl}^0 into a pre-Hilbert space.

Lemma 6.5 (Compatibility with finite-level inner products). *For each $n \geq 0$ and $f \in X_n^{0,\bullet}$ identified with its image in $X_{\text{cyl}}^{0,\bullet}$ via the embedding $j_{n,\infty}^0$ (zero-extension to $V_\infty^\bullet$), $\|f\|_{X_n^0} = \|f\|_{X_{\text{cyl}}^0}$.*

Proof. $\|f\|_{X_n^0}^2 = \sum_{v \in V(G_n^\bullet)} \|f(v)\|_{F^0}^2$; $\|f\|_{X_{\text{cyl}}^0}^2 = \sum_{v \in V_\infty^\bullet} \|f(v)\|^2$, where $f(v) = 0$ for $v \in V_\infty^\bullet \setminus V(G_n^\bullet)$. The two sums agree. $\square$

7. THE INVERSE-LIMIT HILBERT SPACE

Definition 7.1 (Inverse-limit space). The *vertex inverse-limit space* is

$$X_{\text{inv}}^{0,\bullet} := \{(f_n)_{n \geq 0} : f_n \in X_n^{0,\bullet}, p_{n+1,n}^0 f_{n+1} = f_n \text{ for all } n, \sup_n \|f_n\|_{X_n^0} < \infty\}.$$

It is equipped with the inner product

$$\langle f, g \rangle_{X_{\text{inv}}^0} := \lim_{n \rightarrow \infty} \langle f_n, g_n \rangle_{X_n^0}.$$

Lemma 7.2 (Limits exist). *For every bounded compatible family $(f_n)_n \in X_{\text{inv}}^{0,\bullet}$, the sequence $\|f_n\|_{X_n^0}$ is non-decreasing and bounded, hence converges. By polarization, $\langle f_n, g_n \rangle_{X_n^0}$ also converges for any pair of compatible families $(f_n), (g_n)$.*

Proof. By Definition 5.1, $p_{n+1,n}^0$ is restriction to $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$. So the compatibility $f_n = p_{n+1,n}^0 f_{n+1}$ means f_n is the restriction of f_{n+1} to the smaller domain $V(G_n^\bullet)$. Hence $\|f_n\|^2 = \sum_{v \in V(G_n^\bullet)} \|f_{n+1}(v)\|^2 \leq \sum_{w \in V(G_{n+1}^\bullet)} \|f_{n+1}(w)\|^2 = \|f_{n+1}\|^2$. The sequence is non-decreasing. By the supremum bound it is bounded, hence converges. Polarization for cross terms: $\langle f_n, g_n \rangle = \frac{1}{4}(\|f_n + g_n\|^2 - \|f_n - g_n\|^2)$, both norms converge, so does the inner product. $\square$

Theorem 7.3 ($X_{\text{inv}}^0 \cong \ell^2(V_\infty^\bullet)$). *The map*

$$\Phi: X_{\text{inv}}^{0,\bullet} \rightarrow \ell^2(V_\infty^\bullet; F^0), \quad \Phi((f_n)_n) := f_\infty,$$

where $f_\infty(v) := f_n(v)$ for $v \in V(G_n^\bullet)$ (using the compatibility $f_n = p_{n+1,n}^0 f_{n+1}$ to ensure well-definedness across levels), is an $\mathbb{R}$ -linear bijection that intertwines the inner products:

$$\langle f, g \rangle_{X_{\text{inv}}^0} = \langle \Phi(f), \Phi(g) \rangle_{\ell^2(V_\infty^\bullet; F^0)}.$$

In particular, $X_{\text{inv}}^{0,\bullet}$ is a separable Hilbert space.

Proof. *Well-definedness of Φ .* For $v \in V_\infty^\bullet = \bigcup_n V(G_n^\bullet)$, choose n minimal such that $v \in V(G_n^\bullet)$ and set $f_\infty(v) := f_n(v)$. The compatibility $f_n = p_{n+1,n}^0 f_{n+1}$ gives $f_n(v) = f_{n+1}(v)$ for $v \in V(G_n^\bullet)$ at any larger level, so the value is independent of the choice of n .

$f_\infty \in \ell^2(V_\infty^\bullet; F^0)$: by monotone convergence over the increasing finite exhaustion $V(G_n^\bullet) \uparrow V_\infty^\bullet$ together with Lemma 7.2, $\sum_{v \in V_\infty^\bullet} \|f_\infty(v)\|^2 = \lim_n \sum_{v \in V(G_n^\bullet)} \|f_n(v)\|^2 = \lim_n \|f_n\|^2 < \infty$.

Injectivity. If $\Phi(f) = 0$, then $f_\infty(v) = 0$ for all v , so $f_n(v) = 0$ for all $n, v \in V(G_n^\bullet)$, hence $f_n = 0$ for all n .

Surjectivity. For $\phi \in \ell^2(V_\infty^\bullet; F^0)$, define $f_n := \phi|_{V(G_n^\bullet)} \in X_n^0$. Then $\|f_n\|^2 = \sum_{v \in V(G_n^\bullet)} \|\phi(v)\|^2 \leq \|\phi\|_{\ell^2}^2$, bounded. Compatibility: $p_{n+1,n}^0 f_{n+1} = (\phi|_{V(G_{n+1}^\bullet)})|_{V(G_n^\bullet)} = \phi|_{V(G_n^\bullet)} = f_n$. So $(f_n) \in X_{\text{inv}}$ and $\Phi(f) = \phi$.

Inner product preservation. By polarization over the real Hilbert structure (Lemma 7.2) and absolute convergence of the cross-term series via Cauchy-Schwarz on $\ell^2(V_\infty^\bullet; F^0)$, $\langle f, g \rangle_{X_{\text{inv}}} = \lim_n \sum_{v \in V(G_n^\bullet)} \langle f_n(v), g_n(v) \rangle = \sum_{v \in V_\infty^\bullet} \langle \Phi(f)(v), \Phi(g)(v) \rangle = \langle \Phi(f), \Phi(g) \rangle_{\ell^2}$, where the second equality uses monotone convergence on the increasing finite exhaustion $V(G_n^\bullet) \uparrow V_\infty^\bullet$ for each of the polarization summands.

Hilbert + separable. ℓ^2 on a countable set with finite-dimensional fibre is a standard separable Hilbert space [7, Ch. II.4]. The bijection Φ transports this structure to $X_{\text{inv}}^{0,\bullet}$. $\square$

7.1. Density of X_{cyl} in X_{inv} .

Proposition 7.4 (X_{cyl}^0 is dense in X_{inv}^0). *Under the canonical map*

$$X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0) \hookrightarrow \ell^2(V_\infty^\bullet; F^0) \cong X_{\text{inv}}^{0,\bullet},$$

the cylindrical algebraic direct-limit space embeds as a dense subspace of the inverse-limit Hilbert space.

Proof. c_{00} is dense in ℓ^2 on any countable index set, by truncation: for $\phi \in \ell^2(V_\infty^\bullet)$ and $\varepsilon > 0$, choose n large enough that $\sum_{v \in V_\infty^\bullet \setminus V(G_n^\bullet)} \|\phi(v)\|^2 < \varepsilon^2$, then $\phi_n := \phi|_{V(G_n^\bullet)}$ extended by zero is in c_{00} and satisfies $\|\phi - \phi_n\|^2 < \varepsilon^2$. By Theorem 7.3, the same density holds in $X_{\text{inv}}^{0,\bullet}$. $\square$

7.2. Edge sector.

Remark 7.5 (Edge inverse limit: not constructed in this paper). The edge bonding map $p_{n+1,n}^1$ is parent-fibre averaging (not restriction), and satisfies the contraction bound $\|p_{n+1,n}^1\| \leq 1$ (Theorem 5.3(2)). The set of bounded compatible families $\{(A_n) : p_{n+1,n}^1 A_{n+1} = A_n, \sup_n \|A_n\| < \infty\}$ with the induced sup-norm is by the standard coordinatewise-Cauchy argument complete as a Banach space. We do *not*, however, identify this set with a concrete ℓ^2 -type Hilbert

space in this paper: the vertex proof of Theorem 7.3 uses the explicit identification with $\ell^2(V_\infty^\bullet; F^0)$ via restriction-bonding maps, and the averaging-bonding edge sector has no analogous ℓ^2 -identification at present. No edge inverse-limit Hilbert space is constructed below or used downstream in this paper; finite-level edge intertwinings (Theorem 8.5(2)) are all the edge structure we need.

8. COMMUTING SYMMETRY ACTIONS

8.1. The three actions on X_n .

Definition 8.1 (Coxeter action, branchwise). For $g \in W^\bullet$ and $f \in X_n^{0,\bullet}$, define $(g \cdot f)(v) := \rho_{F^0}^\bullet(g) \cdot f(g^{-1} \cdot v)$ for $v \in V(G_n^\bullet)$, where the fibre representation $\rho_{F^0}^\bullet$ depends on the branch:

- On the H_4 branch ($\bullet = H_4$, $W^\bullet = W(H_4)$): $\rho_{F^0}^{H_4}(g)$ acts on $V_{H_4} \cong \mathbb{H} \cong \mathbb{R}^4$ by the standard H_4 -reflection representation associated to the V_{600} icosian root system of [2, Definitions `def:icosian-ring`, `def:V600`] (matrix entries in $\mathbb{Q}(\varphi)$ in the standard quaternion coordinate basis; see also [9, Ch. 2] for the abstract Coxeter group), and trivially on V_{D_4}, V_{16}, V_8 .
- On the D_4 branch ($\bullet = D_4$, $W^\bullet = W(D_4)$): $\rho_{F^0}^{D_4}(g)$ acts on $V_{D_4} \cong \mathbb{R}^4$ by the standard D_4 -reflection representation transported from the abstract D_4 root system through the orthogonal identification $R \in O(4)$ that realizes V_{24} as the rescaled D_4 root polytope in [2, Definition `def:V24` and Proposition `prop:24-in-600`]; that is, for $g \in W(D_4)$ acting on the abstract root space by an even signed-permutation matrix g_0 in $\text{Mat}_4(\mathbb{Z})$ ([9, Ch. 2]), define

$$\rho_{F^0}^{D_4}(g) := R g_0 R^{-1} \in O(V_{D_4}).$$

Writing $R = \frac{1}{\sqrt{2}}H$ where $H \in \text{Mat}_4(\mathbb{Z})$ is the integer matrix from [2, Proposition `prop:24-in-600`] (so $H = \text{diag}(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}, \begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix})$ and $HH^T = 2I_4$), the conjugation is

$$R g_0 R^{-1} = \frac{1}{\sqrt{2}}H \cdot g_0 \cdot \sqrt{2}H^{-1} = H g_0 H^{-1} = \frac{1}{2} H g_0 H^T,$$

using $H^{-1} = \frac{1}{2}H^T$ (since $HH^T = 2I_4$). The product $H g_0 H^T$ has entries in $\mathbb{Z}$, hence $\rho_{F^0}^{D_4}(g) \in \frac{1}{2}\mathbb{Z}^{4 \times 4} \subset \mathbb{Q}^{4 \times 4}$. In particular, $\rho_{F^0}^{D_4}(g)$ preserves the $\mathbb{Q}$ -form $V_{D_4, \mathbb{Q}} = \mathbb{Q}^4$ for every $g \in W(D_4)$. The representation acts trivially on V_{H_4}, V_{16}, V_8 .

The analogous action on $X_n^{1,\bullet}$ acts trivially on $F^1 = \text{Im}(\mathbb{O})$ on both branches and on the edge index by $W^\bullet$ -permutation.

Lemma 8.2 (Unitarity of the Coxeter action). *Under the ℓ^2 -inner products of Definitions 4.1 and 4.2, the Coxeter action of Definition 8.1 is unitary on $X_n^{0,\bullet}$ and on $X_n^{1,\bullet}$.*

Proof. The H_4 - (resp. D_4 -) reflection representation on V_{H_4} (resp. V_{D_4}) is the standard finite real reflection representation of the corresponding Coxeter group, hence orthogonal with respect to the Euclidean metric on the fibre by the standard theorem on finite real reflection groups [9, Ch. 1]; the underlying root system data is fixed in P2 ([2, Definition `def:V600`, Definition `def:V24` and Proposition `prop:24-in-600`]). The representation acts trivially on the other fibre summands; hence $\rho_{F^0}^\bullet(g)$ is a fibre isometry. The vertex (resp. oriented-edge) measure $\mu_n^\bullet$ (resp. $\nu_n^\bullet$) is $W^\bullet$ -invariant by the Coxeter-equivariance of the refinement (Proposition 2.8). The combined pre-composition with g^{-1} and fibrewise action by $\rho_{F^0}^\bullet(g)$ is therefore a unitary on the ℓ^2 -space. $\square$

8.2. $\mathbb{Q}$ -form and the σ -action.

Definition 8.3 (Mixed $\mathbb{Q}(\varphi)/\mathbb{Q}$ -form of the vertex sector). By inspection of the explicit fibre constructions in [2], and in particular the standard sector $\mathbb{Q}$ -forms collected in [2, Definition `def:Q-forms`] (named there $U_{\mathbb{Q}} = \mathbb{Q}^4$, $\text{Cl}_{\mathbb{Q}}(1, 3)$, $\mathbb{O}_{\mathbb{Q}}$, which we re-use as $V_{D_4, \mathbb{Q}}$, $V_{16, \mathbb{Q}}$, $V_{8, \mathbb{Q}}$ respectively): $V_{D_4} = \mathbb{R}^4$ has the standard $\mathbb{Q}$ -form $V_{D_4, \mathbb{Q}} := \mathbb{Q}^4$ ([2, Definition `def:V24` and Proposition `prop:24-in-600`]); $V_{16} = \text{Cl}(1, 3)$ has the $\mathbb{Q}$ -form $V_{16, \mathbb{Q}} := \mathbb{Q}\{\gamma_A\}$ (rational span of the Clifford monomial basis, [2, Definition `def:Cl-basis`], which is a $\mathbb{Q}$ -basis of a

$\mathbb{Q}$ -form whose $\mathbb{R}$ -extension equals $\text{Cl}(1, 3)$); $V_8 = \mathbb{O}$ has the $\mathbb{Q}$ -form $V_{8,\mathbb{Q}} := \mathbb{Q}\{1, e_1, \dots, e_7\}$ (rational span of the standard octonion basis, [2, Definition `def:octonions`]). For $V_{H_4} = \mathbb{R}^4$ the natural rational structure is the $\mathbb{Q}(\varphi)$ -form $V_{H_4,\mathbb{Q}(\varphi)} := \mathbb{Q}(\varphi)^4 \subset \mathbb{R}^4$ via the standard real embedding $\iota: \mathbb{Q}(\varphi) \hookrightarrow \mathbb{R}$ sending φ to the golden ratio $\frac{1}{2}(1 + \sqrt{5})$, because the icosian arithmetic on V_{H_4} is naturally over $\mathbb{Z}[\varphi]$ rather than over $\mathbb{Z}$ ([2, Definition `def:icosian-ring` and Theorem `thm:icosian-closure`]).

The combined rationality data is the *mixed* $\mathbb{Q}(\varphi)/\mathbb{Q}$ -form

$$F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0 := V_{H_4,\mathbb{Q}(\varphi)} \oplus V_{D_4,\mathbb{Q}} \oplus V_{16,\mathbb{Q}} \oplus V_{8,\mathbb{Q}},$$

which is a $\mathbb{Q}$ -vector space of total $\mathbb{Q}$ -dimension $8 + 4 + 16 + 8 = 36$ (since $V_{H_4,\mathbb{Q}(\varphi)} = \mathbb{Q}(\varphi)^4$ has $\mathbb{Q}$ -dimension 8). Its $\mathbb{R}$ -extension is sector-dependent: on the H_4 sector we use $V_{H_4,\mathbb{Q}(\varphi)} \otimes_{\mathbb{Q}(\varphi)} \mathbb{R} = \mathbb{R}^4 = V_{H_4}$ via ι , on the other three sectors we use $V_{*,\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R} = V_*$. With this convention

$$F^0 = V_{H_4} \oplus V_{D_4} \oplus V_{16} \oplus V_8 \quad (\dim_{\mathbb{R}} F^0 = 32)$$

is recovered from $F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0$. We do *not* claim $F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0 \otimes_{\mathbb{Q}} \mathbb{R} = F^0$; that would have $\mathbb{R}$ -dimension 36, not 32, because $\mathbb{Q}(\varphi) \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R} \oplus \mathbb{R}$ (the two real roots of $x^2 - x - 1$). The $\mathbb{R}$ -dimension 32 of F^0 comes from the $\mathbb{Q}(\varphi)$ -tensor product on the H_4 sector, not from the $\mathbb{Q}$ -tensor product.

Define the mixed-form vertex space at level n as the $\mathbb{Q}$ -vector space of all $F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0$ -valued functions on the finite set $V(G_n^\bullet)$,

$$X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}} := \{f: V(G_n^\bullet) \rightarrow F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0\}.$$

(No completion is needed because the indexing set is finite; we do not use the $\mathbb{Q}$ -vector-space structure as a Hilbert space in this paper.)

Definition 8.4 (σ -action on the mixed-form vertex sector). The Galois involution $\sigma: \mathbb{Q}(\varphi) \rightarrow \mathbb{Q}(\varphi)$ defined by $\sigma(\varphi) = 1 - \varphi$ ([1, Def. `def:sigma`]) extends sector-by-sector to a $\mathbb{Q}$ -linear involution $\sigma_n: X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \rightarrow X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ on the mixed-form vertex space of Definition 8.3 as follows:

- on the $V_{H_4,\mathbb{Q}(\varphi)} = \mathbb{Q}(\varphi)^4$ component, σ_n acts coefficientwise by σ in each of the four $\mathbb{Q}(\varphi)$ -coordinates;
- on each of $V_{D_4,\mathbb{Q}}, V_{16,\mathbb{Q}}, V_{8,\mathbb{Q}}$, σ_n acts as the identity (these sectors carry no $\mathbb{Q}(\varphi)$ -rationality).

σ_n is a $\mathbb{Q}$ -linear involution on the $\mathbb{Q}$ -vector space $X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$. It does *not* extend to an $\mathbb{R}$ -linear involution on the $\mathbb{R}$ -extension $X_n^{0,\bullet} = \ell^2(V(G_n^\bullet), \mu_n^\bullet; F^0)$ through the standard real embedding ι (because σ on $\mathbb{Q}(\varphi) \otimes_{\mathbb{Q}(\varphi),\iota} \mathbb{R} = \mathbb{R}$ is fixed by ι but free over the conjugate embedding $\bar{\iota}$ sending φ to $1 - \varphi$); σ_n is the *rationality-side* involution, used in the intertwining identities below at the mixed-form level.

We do not define a σ -action on the edge sector X_n^1 in this paper: the edge fibre $\text{Im}(\mathbb{O})$ has no canonical $\mathbb{Q}(\varphi)$ -coefficient structure relevant to the cascade arithmetic, and downstream papers needing edge σ -actions construct them separately.

8.3. The intertwining theorem.

Theorem 8.5 (Intertwining identities). *For every $n \geq 0$ and $\bullet \in \{H_4, D_4\}$:*

- (1) (*Refinement-Coxeter, vertex sector.*) *For every $g \in W^\bullet$,*

$$g \circ p_{n+1,n}^0 = p_{n+1,n}^0 \circ g \quad \text{on } X_{n+1}^{0,\bullet},$$

and analogously $g \circ j_{n,n+1}^0 = j_{n,n+1}^0 \circ g$.

- (2) (*Refinement-Coxeter, edge sector.*) *The same identities hold for $p_{n+1,n}^1$ on $X_{n+1}^{1,\bullet}$ and for $j_{n,n+1}^1$ on $X_n^{1,\bullet}$, i.e. $g \circ p_{n+1,n}^1 = p_{n+1,n}^1 \circ g$ and $g \circ j_{n,n+1}^1 = j_{n,n+1}^1 \circ g$.*

- (3) (*Refinement- σ , vertex sector, mixed-form level.*) *On the mixed-form vertex space $X_{n+1}^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ of Definition 8.3,*

$$\sigma_n \circ p_{n+1,n}^0|_{\mathbb{Q}(\varphi)/\mathbb{Q}} = p_{n+1,n}^0|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \circ \sigma_{n+1},$$

where $p_{n+1,n}^0|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ is the mixed-form restriction of the bonding map (well-defined because vertex restriction preserves the sectorwise rationality) and σ_n, σ_{n+1} are the involutions of Definition 8.4. Analogously, with $j_{n,n+1}^0$ the (zero-extension) adjoint of $p_{n+1,n}^0$ from Proposition 5.5,

$$\sigma_{n+1} \circ j_{n,n+1}^0|_{\mathbb{Q}(\varphi)/\mathbb{Q}} = j_{n,n+1}^0|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \circ \sigma_n$$

on $X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ (the σ indices swap relative to the p^0 identity since j goes upward $n \rightarrow n+1$ while p goes downward).

- (4) (Sectorwise Coxeter- σ , vertex sector.) Let $g|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ denote the action of $g \in W^\bullet$ on the mixed-form vertex space $X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$, defined by the same formula as in Definition 8.1 when the fibrewise reflection $\rho_{F^0}^\bullet(g)$ preserves $F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0$ (so the resulting function is again $F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0$ -valued); we make the well-definedness case-explicit below.

- (i) (D_4 branch.) For $\bullet = D_4$ and any $g \in W(D_4)$, $\rho_{F^0}^{D_4}(g)$ has matrix entries in $\mathbb{Q}$ (in the ambient $V_{D_4} = \mathbb{R}^4$ basis used here), as established in Definition 8.1, hence preserves the $\mathbb{Q}$ -form $V_{D_4,\mathbb{Q}} = \mathbb{Q}^4$. It acts trivially on V_{H_4}, V_{16}, V_8 . Therefore $g|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ is well-defined for all $g \in W(D_4)$, and

$$g|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \circ \sigma_n = \sigma_n \circ g|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \quad \text{on } X_n^{0,D_4}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}.$$

- (ii) (H_4 branch, restricted subspace.) For $\bullet = H_4$ let $X_n^{0,H_4}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}^b \subset X_n^{0,H_4}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ denote the subspace of functions $V(G_n^{H_4}) \rightarrow V_{D_4,\mathbb{Q}} \oplus V_{16,\mathbb{Q}} \oplus V_{8,\mathbb{Q}}$ (i.e. valued in the σ -trivial sectors only). For all $g \in W(H_4)$, $\rho_{F^0}^{H_4}(g)$ acts trivially on $V_{D_4} \oplus V_{16} \oplus V_8$ (Definition 8.1), so $g|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ is well-defined on $X_n^{0,H_4}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}^b$, and

$$g|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \circ \sigma_n = \sigma_n \circ g|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \quad \text{on } X_n^{0,H_4}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}^b.$$

- (iii) (H_4 branch, V_{H_4} summand: not claimed.) On the complementary V_{H_4} summand of $X_n^{0,H_4}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$, the commutation $g|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \circ \sigma_n = \sigma_n \circ g|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ is not delivered by [1, Theorem `thm:pisigma-functorial`] in its forward direction (since not all elements of $W(H_4)$ are scalar extensions of $\mathbb{Q}$ -linear maps on the chosen standard $\mathbb{Q}^4 \subset \mathbb{Q}(\varphi)^4$); we do not claim it here. See Remark 8.6.

The vertex Coxeter intertwining of (1) descends to the vertex cylindrical and inverse-limit spaces $X_{\text{cyl}}^{0,\bullet}, X_{\text{inv}}^{0,\bullet}$ by linearity and continuity (using Theorem 7.3 for the inverse limit). The edge Coxeter intertwining of (2) is stated at the finite level only; we do not construct an edge inverse-limit Hilbert space in this paper. The σ -intertwining of (3) is stated only at the mixed-form level $X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$; we do not claim a σ -action on the $\mathbb{R}$ -Hilbert spaces $X_n^{0,\bullet}, X_{\text{cyl}}^{0,\bullet}, X_{\text{inv}}^{0,\bullet}$, since (per Definition 8.4) σ does not extend $\mathbb{R}$ -linearly through the standard real embedding ι .

Remark 8.6 (Why Coxeter- σ commutation is restricted). The forward direction of [1, Theorem `thm:pisigma-functorial`] applies here only when the map under consideration is the scalar extension T_K of a $\mathbb{Q}$ -linear endomorphism $T: V \rightarrow V$ of an underlying $\mathbb{Q}$ -vector space V (so T stabilises the fixed inclusion $V \otimes 1 \subset V \otimes_{\mathbb{Q}} \mathbb{Q}(\varphi)$). On the $V_{H_4,\mathbb{Q}(\varphi)}$ component, the $W(H_4)$ action is given by reflection matrices whose coefficients lie in $\mathbb{Q}(\varphi)$ (not in $\mathbb{Q}$), because the icosian roots have $\mathbb{Q}(\varphi)$ -rational coordinates ([2, Definitions `def:V600`, `def:icosian-ring`]). Such a reflection is $\mathbb{Q}(\varphi)$ -linear (hence a fortiori $\mathbb{Q}$ -linear on the underlying $\mathbb{Q}$ -vector space $V_{H_4,\mathbb{Q}(\varphi)}$), but it does *not* restrict to a $\mathbb{Q}$ -linear endomorphism of the chosen standard $\mathbb{Q}$ -form $\mathbb{Q}^4 \subset \mathbb{Q}(\varphi)^4$ (the σ -fixed sublattice with respect to the basis fixed by [2, Definition `def:V600`]): a non-trivial $\mathbb{Q}(\varphi)$ -coefficient reflection of a $\mathbb{Q}^4$ -vector carries it out of $\mathbb{Q}^4$ into the general $\mathbb{Q}(\varphi)^4$ space. The forward-direction hypothesis of [1, Theorem `thm:pisigma-functorial`] is therefore not satisfied for the chosen standard $\mathbb{Q}$ -form, so the source theorem does not deliver $g \circ \sigma_n = \sigma_n \circ g$ on the V_{H_4} sector. We do not assert in this paper that the equality fails; we assert only that it is not delivered automatically by the source theorem and is not claimed here.

The sectorwise statement (4) records the cases where commutation *does* hold automatically under the branchwise action of Definition 8.1: the entire D_4 -branch action (since $W(D_4)$ acts only on V_{D_4} , which is σ -trivial), and the H_4 -branch action restricted to the σ -trivial summands $V_{D_4, \mathbb{Q}}, V_{16, \mathbb{Q}}, V_{8, \mathbb{Q}}$ (on which $W(H_4)$ acts trivially).

For downstream papers needing the full $W(H_4)$ - σ relationship (e.g. in spectral-arithmetic constructions), a natural formulation is as an intertwining with σ -conjugate Coxeter elements g^σ rather than as commutation: $\sigma \circ g = g^\sigma \circ \sigma$ where g^σ is the matrix-coefficient σ -conjugate of g .

Proof. (1) Refinement-Coxeter, vertex. For $f \in X_{n+1}^0, v \in V(G_n^\bullet), g \in W^\bullet$:

$$\begin{aligned} (g \circ p_{n+1, n}^0 f)(v) &= \rho_{F^0}(g) \cdot (p_{n+1, n}^0 f)(g^{-1} \cdot v) = \rho_{F^0}(g) \cdot f(g^{-1} \cdot v) \\ &= (g \cdot f)(v) = (p_{n+1, n}^0 (g \cdot f))(v). \end{aligned}$$

This uses $g^{-1} \cdot v \in V(G_n^\bullet)$ (because $V(G_n^\bullet)$ is $W^\bullet$ -stable) and the equality $f(g^{-1}v) = (p_{n+1, n}^0 f)(g^{-1}v)$ (definition of p^0 as restriction).

The j -version follows by adjunction: $g \circ j = j \circ g$ is equivalent to $g^{-1} \circ p = p \circ g^{-1}$ (which we just proved with g^{-1} in place of g).

(2) *Refinement-Coxeter, edge.* For $A \in X_{n+1}^1, e \in E^{\text{or}}(G_n^\bullet), g \in W^\bullet$ (with the $W^\bullet$ action on $\text{Im}(\mathbb{O})$ trivial, so we drop the $\rho_{F^1}(g)$ factor):

$$(g \circ p_{n+1, n}^1 A)(e) = (p_{n+1, n}^1 A)(g^{-1} \cdot e) = \frac{1}{2} \sum_{e' \in (\pi^E)^{-1}(g^{-1}e)} A(e').$$

By Proposition 2.8(3), $(\pi^E)^{-1}(g^{-1}e) = g^{-1} \cdot (\pi^E)^{-1}(e)$. Substituting $e'' := g \cdot e'$:

$$= \frac{1}{2} \sum_{e'' \in (\pi^E)^{-1}(e)} A(g^{-1} \cdot e'') = \frac{1}{2} \sum_{e''} (g \cdot A)(e'') = (p_{n+1, n}^1 (g \cdot A))(e).$$

The corresponding identity for the adjoint $j_{n, n+1}^1 = (p_{n+1, n}^1)^*$ follows by adjunction together with the unitarity of the $W^\bullet$ -action (Lemma 8.2): for any $A \in X_n^1, B \in X_{n+1}^1, g \in W^\bullet$,

$$\begin{aligned} \langle j_{n, n+1}^1 (g \cdot A), B \rangle &= \langle g \cdot A, p_{n+1, n}^1 B \rangle = \langle A, g^{-1} \cdot p_{n+1, n}^1 B \rangle = \langle A, p_{n+1, n}^1 (g^{-1} \cdot B) \rangle \\ &= \langle j_{n, n+1}^1 A, g^{-1} \cdot B \rangle = \langle g \cdot j_{n, n+1}^1 A, B \rangle, \end{aligned}$$

where the third equality is the just-proved p^1 -identity. Since B is arbitrary, $j_{n, n+1}^1 (g \cdot A) = g \cdot j_{n, n+1}^1 A$, i.e. $g \circ j_{n, n+1}^1 = j_{n, n+1}^1 \circ g$.

(3) *Refinement- σ .* The bonding map $p_{n+1, n}^0$ is vertex-restriction along $V_n^\bullet \subset V_{n+1}^\bullet$ and acts componentwise on F^0 . Restricted to the mixed-form $X_{n+1}^{0, \bullet} |_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ (Definition 8.3), it produces a function on $V_n^\bullet$ valued in $F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0$, hence a well-defined map $p_{n+1, n}^0 |_{\mathbb{Q}(\varphi)/\mathbb{Q}}: X_{n+1}^{0, \bullet} |_{\mathbb{Q}(\varphi)/\mathbb{Q}} \rightarrow X_n^{0, \bullet} |_{\mathbb{Q}(\varphi)/\mathbb{Q}}$. By Definition 8.4, σ_n acts on $F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0$ pointwise (sectorwise), and vertex-restriction commutes pointwise with this fibrewise action. The identity $\sigma_n \circ p_{n+1, n}^0 |_{\mathbb{Q}(\varphi)/\mathbb{Q}} = p_{n+1, n}^0 |_{\mathbb{Q}(\varphi)/\mathbb{Q}} \circ \sigma_{n+1}$ therefore follows by direct fibrewise inspection at each vertex $v \in V_n^\bullet$.

For the adjoint $j_{n, n+1}^0$ (zero-extension on $V_{n+1}^\bullet \setminus V_n^\bullet$), the same fibrewise inspection together with σ acting trivially on the zero fibre gives $\sigma_{n+1} \circ j_{n, n+1}^0 |_{\mathbb{Q}(\varphi)/\mathbb{Q}} = j_{n, n+1}^0 |_{\mathbb{Q}(\varphi)/\mathbb{Q}} \circ \sigma_n$, with the σ indices reversed because j goes upward.

(4) *Sectorwise Coxeter- σ .* On the D_4 branch, $W(D_4)$ acts (by Definition 8.1) on V_{D_4} and trivially on V_{H_4}, V_{16}, V_8 ; since σ acts trivially on $V_{D_4, \mathbb{Q}}$ and on $V_{16, \mathbb{Q}}, V_{8, \mathbb{Q}}$, the composition $g \circ \sigma$ equals $\sigma \circ g$ on every fibre summand, so it equals $\sigma \circ g$ on $X_n^{0, D_4} |_{\mathbb{Q}(\varphi)/\mathbb{Q}}$. On the H_4 branch, $W(H_4)$ acts on V_{H_4} and trivially on V_{D_4}, V_{16}, V_8 ; on the latter three summands σ also acts trivially, so the commutation is automatic there by the same fibrewise computation. The remaining V_{H_4} summand is excluded from (4) (Remark 8.6).

The descent to $X_{\text{cyl}}^{0, \bullet}$ and $X_{\text{inv}}^{0, \bullet}$ applies to the vertex Coxeter action of (1) only. That action is continuous on each $X_n^{0, \bullet}$, and the intertwining $g \circ p_{n+1, n}^0 = p_{n+1, n}^0 \circ g$ is preserved under the algebraic direct-limit construction (Theorem 6.2) and the ℓ^2 -completion (Theorem 7.3). The

edge Coxeter action of (2) is stated only at the finite level since this paper does not construct an edge inverse-limit Hilbert space; the refinement- σ identities of (3) are stated only at the finite mixed-form level $X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ since (per Definition 8.4) σ does not extend $\mathbb{R}$ -linearly to $X_n^{0,\bullet}$, $X_{\text{cyl}}^{0,\bullet}$, or $X_{\text{inv}}^{0,\bullet}$. $\square$

9. CITATION CONTRACT AND DOWNSTREAM HANDOFF

9.1. What downstream papers may cite.

- (1) Definitions 4.1, 4.2: finite-level Hilbert sector spaces $X_n^{0,\bullet}, X_n^{1,\bullet}$ under counting measure.
- (2) Theorem 5.3: contractive downward bonding maps $p_{n+1,n}^0$ (restriction) and $p_{n+1,n}^1$ (parent-fibre averaging).
- (3) Proposition 5.5: cylindrical embeddings $j_{n,n+1}^0, j_{n,n+1}^1$ as adjoints of the bonding maps, with the section relations $p_{n+1,n}^0 \circ j_{n,n+1}^0 = \text{id}_{X_n^0}$ (vertex sector) and $p_{n+1,n}^1 \circ j_{n,n+1}^1 = \frac{1}{2} \text{id}_{X_n^1}$ (edge sector); see also Remark 5.6 for the unscaled section s^1 in the edge sector.
- (4) Theorem 6.2: $X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0)$ as algebraic direct limit (vertex sector).
- (5) Theorem 7.3: $X_{\text{inv}}^{0,\bullet} \cong \ell^2(V_\infty^\bullet; F^0)$ as separable Hilbert space (vertex sector).
- (6) Proposition 7.4: $X_{\text{cyl}}^{0,\bullet}$ dense in $X_{\text{inv}}^{0,\bullet}$.
- (7) Theorem 8.5: refinement-Coxeter (vertex finite-level and inverse-limit; edge finite-level only), refinement- σ (mixed-form vertex sector only, on both p^0 and j^0), and Coxeter- σ (restricted, see Remark 8.6) intertwining identities.

9.2. Recommended citation phrasing. “By P3 [3], the cascade vertex substrate is $\ell^2(V_\infty^\bullet; F^0)$ as the projective inverse limit of $(X_n^{0,\bullet}, p_{n+1,n}^0)_n$ with $p_{n+1,n}^0$ restriction (Theorems 5.3, 7.3); the cylindrical direct limit $X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0)$ embeds densely in $X_{\text{inv}}^{0,\bullet}$ (Theorem 6.2, Proposition 7.4); the refinement-Coxeter intertwining holds for the vertex sector at every finite level and on the inverse limit, and for the edge sector at every finite level (no edge inverse limit is constructed); the refinement- σ intertwining holds for both p^0 and j^0 on the mixed-form vertex sector $X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$; and the Coxeter- σ commutation holds in the restricted sense of Theorem 8.5(4) and Remark 8.6 (full $W(H_4)$ - σ commutation on the V_{H_4} summand is not supplied by [1, Theorem thm:pisigma-functorial] in its forward direction and is not asserted here; some $W(H_4)$ reflections have non-rational $\mathbb{Q}(\varphi)$ -coefficient matrix entries that fall outside that theorem’s $\mathbb{Q}$ -form hypothesis when applied to the chosen standard $\mathbb{Q}^4 \subset \mathbb{Q}(\varphi)^4$.”

9.3. What this paper does not provide. No closure-functional / dynamics theorem (P4); no metric, gauge, spectral, Hecke, σ -cohomology, or computation correspondence (P5–P13); no continuum-limit identification of X_{inv} with any classical L^2 space; no physical interpretation.

APPENDIX A. NOTATION AUDIT

Symbol	Meaning	Defined
$\bullet$	branch decoration (H_4 or D_4)	Def. 2.1
V_{600}, V_{24}	base vertex sets (P2)	Def. 2.1
$G_n^\bullet$	graph at refinement level n	Def. 2.3
$V(G_n^\bullet)$	vertex set, with $V_n \subset V_{n+1}$	Def. 2.3
$M_n^\bullet, C_n^\bullet$	midpoints and centroids at level n	Def. 2.3
$\pi_{n+1,n}^E$	canonical edge-parent map	Def. 2.6
$\mu_n^\bullet, \nu_n^\bullet$	vertex / edge counting measures	Def. 3.1
F^0	vertex fibre $V_{H_4} \oplus V_{D_4} \oplus V_{16} \oplus V_8$	Notation 3.3
F^1	edge fibre $\text{Im}(\mathbb{O})$ (fixed)	Notation 3.3
X_n^0, X_n^1	finite-level ℓ^2 spaces	Defs. 4.1, 4.2
$p_{n+1,n}^0$	vertex bonding (restriction)	Def. 5.1
$p_{n+1,n}^1$	edge bonding (parent-fibre averaging)	Def. 5.2
$j_{n,n+1}^0$	vertex embedding (zero-extension)	Def. 5.4
$j_{n,n+1}^1$	edge embedding (parent-replication)	Def. 5.4
$V_\infty^\bullet$	$\bigcup_n V(G_n^\bullet)$	Thm. 6.2
$X_{\text{cyl}}^{0,\bullet}$	$\cong c_{00}(V_\infty^\bullet; F^0)$	Def. 6.1, Thm. 6.2
$X_{\text{inv}}^{0,\bullet}$	$\cong \ell^2(V_\infty^\bullet; F^0)$	Def. 7.1, Thm. 7.3
σ	coefficientwise Galois involution per P1	Def. 8.4
$W^\bullet$	Coxeter group ($W(H_4)$ or $W(D_4)$)	Def. 2.1

REFERENCES

- [1] *Cascade-Classical Correspondence I: Sigma-Fixed Rationality over $\mathbb{Q}(\varphi)$* , `papers/cascade-sigma-rationality/cascade-sigma-rationality.tex`. (Paper P1.) Pinned results used in this paper, cited by label: Definition `def:sigma` (the Galois involution $\sigma: \mathbb{Q}(\varphi) \rightarrow \mathbb{Q}(\varphi)$); Definition `def:sigmaV` (coefficientwise σ_V on $V \otimes_{\mathbb{Q}} \mathbb{Q}(\varphi)$); Theorem `thm:pisigma-functorial` (functoriality of the scalar extension T_K for a $\mathbb{Q}$ -linear map $T: V \rightarrow V$). We use `thm:pisigma-functorial` only in its forward direction (the scalar extension of a $\mathbb{Q}$ -linear map intertwines σ_V); the converse, that any σ_V -intertwining map descends to a $\mathbb{Q}$ -linear map on a chosen fixed $\mathbb{Q}$ -form, is our local inference, not a statement in P1.
- [2] *Cascade-Classical Correspondence II: The Cascade Algebraic Substrate*, `papers/cascade-algebraic-substrate/cascade-algebraic-substrate.tex`. (Paper P2.) Pinned results used in this paper, cited by label: `def:icosian-ring` (the icosian ring $I \subset \mathbb{H}$ as a rank-8 $\mathbb{Z}$ -subring of the quaternions, generated by the 120 unit icosians, including half-integer combinations; the arithmetic carrier of V_{H_4}); `def:V600` ($V_{600} \subset \mathbb{H}$ vertex set and coordinates, with entries in $\mathbb{Q}(\varphi)$, including halves); `def:V24` and `prop:24-in-600` ($V_{24} \subset V_{600}$ as the rescaled D_4 root polytope); `thm:icosian-closure` (closure of the icosian ring under its multiplication, used here only as a structural reference); `def:Cl-basis` (the $\text{Cl}(1,3)$ monomial basis $\{\gamma_A\}$); `def:octonions` (standard octonion basis); `def:Q-forms` (the standard $\mathbb{Q}$ -forms on V_{D_4}, V_{16}, V_8 used in this paper's mixed $\mathbb{Q}(\varphi)/\mathbb{Q}$ -form construction; P2 names them $U_{\mathbb{Q}}, \text{Cl}_{\mathbb{Q}}(1,3), \mathbb{O}_{\mathbb{Q}}$ respectively). **Inference, not stated theorem in P2:** the standard $W(H_4)$ reflection generators have matrix entries in $\mathbb{Q}(\varphi)$ when written in the icosian basis. P2 supplies the $\mathbb{Q}(\varphi)$ -rational coordinates of the icosian roots (`def:V600, def:icosian-ring`); the matrix-entry statement follows by the standard reflection formula and is used in this paper's H_4 branch.
- [3] (This paper.) *Cascade-Classical Correspondence III: Refinement Spaces and Symmetry Actions*.
- [4] *Cascade Closure Dynamics: Finite Closure Functionals, Gradient Flow, and Refinement Compatibility*, paper P4. `papers/cascade-closure-dynamics/cascade-closure-dynamics.tex`. The downstream paper that puts dynamics on the chain spaces constructed here.
- [5] *The Cascade Curvature Operator under Refinement: Schur-Complement Identity and Energy Intertwining*. `papers/cascade-refinement-compat/cascade-refinement-compat.tex`. Discharges the residual-monotonicity obligation (O3) of [6] for the scaled vertex Laplacian in an abstract scalar pure-midpoint refinement model; §`sec:scope-and-gap` of that paper names the open lift hypotheses (L1) and (L3) needed to extend its results to the full refinement datum constructed here.
- [6] *Cascade Closure: A Conditional Compatibility Template for Geometric State Selection*, paper P-mech. `papers/cascade-mechanism/cascade-mechanism.tex`. Defines the consumer API (O0)–(O3) that [5] discharges in the abstract model.
- [7] M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, rev. ed., Academic Press, 1980. Hilbert spaces and direct/inverse limits: Ch. II; ℓ^2 -spaces on countable sets: Ch. II.4; algebraic direct limits: Ch. II.1.4.

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- [10] H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover, 1973. 24-cell: §22.2; 600-cell coordinates: §22.7; Schläfli refinement: §8.
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- [12] P. Lounesto, *Clifford Algebras and Spinors*, 2nd ed., London Math. Soc. Lecture Note Series **286**, Cambridge University Press, 2001. Hilbert–Schmidt-style trace forms (with the principal involution $a \mapsto a^\dagger$): Ch. 9.